AIP Conference Proceedings

Hydrogen line broadening in Dense Plasmas: Important physics

Cite as: AIP Conference Proceedings **645**, 302 (2002); https://doi.org/10.1063/1.1525469
Published Online: 15 November 2002

S. Alexiou

View Online
Export Citation

AIP | Conference Proceedings
Get 30% off all print proceedings!
Enter Promotion Code PDF30 at checkout

© 2002 American Institute of Physics.

HTML ABSTRACT • LINKS

Hydrogen line broadening in Dense Plasmas: Important physics

S.Alexiou

Abstract.
In view of recent high density experiments on hydrogen lines and highly exaggerated reports of new couplings and effects, this work discusses some *known* physics not normally considered in the analysis of these works that is of much more importance in the dense plasma regime than recently proposed exotic effects.

In the last Spectral Line Shapes Conference[1] convergence of quantal and nonperturbative semiclassical calculations for broadening of isolated ion lines was reported. The convergence was achieved by allowing for penetrating collisions. Further work on a much larger set of isolated ion lines seems to verify this convergence. Note that theory-experiment agreement has always been less than satisfactory for such lines.

In contrast for hydrogen lines theory-experiment agreement has been very good for lines without a central component(such as H_β) ever since accurate quasistatic microfield distributions became available(with the exception of the dip, which needs a good description of the dynamics too) and has been practically perfect for all lines when ion dynamics is taken into account for low and moderate densities. Recently, however, some high density experiments have reported[2, 3, 4] deviations from the standard theory(ST) i.e. perturbative impact theory for electrons and quasistatic ions. In these experiments ions are usually well quasistatic, so the main issue is electron broadening. Although the degree of confidence given to these experiments must be carefully evaluated(at least one seems to have been affected by optical thickness effects[5]), the theoretical calculations should be even more carefully examined, especially as a number of new couplings and effects have been proposed, some of them quite exotic and actually known to be missing important physics and lead to incorrect results. An example is the so-called "Advanced Generalized Theory"(AGT)[6] (p.562):(the present author never saw this paper before it was printed) "However, even for the La line at densities of the order of $10^{17} cm^{-3}$, where the experimental width is by a factor of two greater than the KG width and the entire difference between the two widths was usually "blamed" on the ion dynamics, it turns out the following: The AGT eliminates about one half of this discrepancy just by treating electrons more accurately than in the KG theory, which might mean that the ion-dynamical contribution could be about a factor of two smaller than it was previously thought". This statement is cast in an even stronger form in the abstract of the same source: "...is in reality about a factor of two smaller than it was previously assumed." The truth is that more than a decade before that publication, a number of authors, for instance Hegerfeld and Kesting(HK)[7], had used exact calculations to establish the im-

CP645, *Spectral Line Shapes: Volume 12, 16th ICSLS,* edited by C. A. Back
© 2002 American Institute of Physics 0-7354-0100-4/02/$19.00

portance of ion dynamics for the Grützmacher-Wende experiments(i.e. La at the density range mentioned above). As early as 1988, HK had confirmed by including much more physics and an exact treatment(i.e. solving the Schrödinger equation(SE)) that: a) There is no coupling between electrons and ions for the line and parameters in question, that is the joint electron-ion simulation produces an autocorrelation function that is the product of the autocorrelation functions solved with ions and electrons alone and b) the electronic autocorrelation function is in excellent agreement with the ST. In other words, HK treat ion dynamics exactly, treat electron-ion coupling exactly, go beyond the impact approximation and provide an exact semiclassical dipole solution of the dynamics, while the AGT does none of that, yet still claims that HK is in error[1].

In the present work we concentrate on some physics that is *not* included in either standard or exotic calculations and that is important for the conditions of these experiments. Although it would also be very interesting to access the importance of quantal effects too, such issues are not discussed here.

QUALITATIVE CONSIDERATIONS

The ST for hydrogen emitters gives a collision operator with a logarithmic divergence:

$$\Phi \approx ln(\rho_{max}/\rho_{min}) \tag{1}$$

where ρ_{max} and ρ_{min} are maximum and minimum impact parameters. As long as the argument of the logarithm is large, these cutoffs are not too important. However, for large densities/small temperatures, this ratio becomes small(for instance 1.6 in a case considered below) and cutoffs are critical. In such a case the anwers obtained by different choices of the cutoffs and assumed strong collision contributions can differ significantly. Therefore treating small impact parameters right is important at high densities. One should keep in mind that what physically removes the divergence at small impact parameters(and preserves perturbation theory) is the breakdown of the long range dipole approximation. Therefore taking account of penetration can be critical, i.e. the dipole approximation should be replaced by the full Coulomb interaction:

$$e\mathbf{r}\cdot\mathbf{E}(t) \rightarrow e[|\mathbf{r}-\mathbf{R}(t)|^{-1} - R^{-1}(t)] \tag{2}$$

Although one may remove at least some of the logarithmic divergence for instance by neglecting time ordering or by dressing the electron field(e.g. by the ion+z- component of the electronic field, as in the AGT), this does *not* address the *fundamental* reason for the finite electronic contribution.

---

[1] Theorywise, the AGT, which uses as the unperturbed Hamiltonian the static ion plus electronic field in that direction is an interesting idea; however, the U-matrix elements are connected by the SE and unitarity. The inaccuracy in the computation of one of them will necessarily show in the others and in general it is not possible to compute some parts of the problem inaccurately without affecting all components. This raises the question that if it is important to treat the z-component of the electronic field nonperturbatively, why is it ok to treat the x and y-components perturbatively.

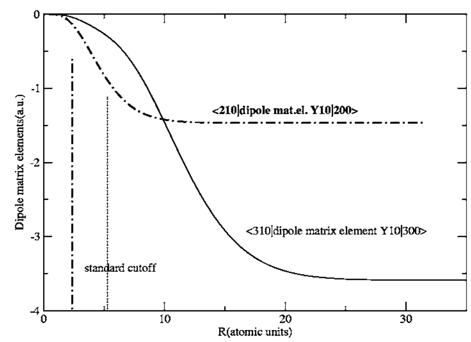

FIGURE 1. Onset of penetration vs. standard estimate(vertical lines).

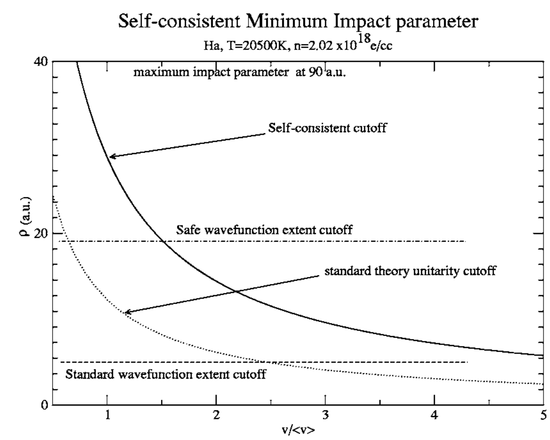

FIGURE 2. Importance of penetration for minimum impact parameter determination.

In [1] it was shown that the standard estimate of the wavefunction extent(WE) $n_u^2 a_0/Z$, with n_u the upper level principal quantum number, a_0 the Bohr radius and Z the spectroscopic charge number, is seriously inaccurate for modern day purposes. Fig.1, which plots dipole matrix elements for L_α and H_α versus perturber position R, shows that things are no different in the case of hydrogen. This means that the contribution from small impact parameters for which penetration is important, is much smaller than the standard predictions and that the WE cutoff employed to preserve the dipole approximation(i.e. $n_u^2 a_0/Z$) should be revised.

To illustrate the importance of these issues, consider the H_α measurement in [4]. In Fig.2 we plot the ST unitarity-based cutoff(dotted line) $(n_u^2 - n_l^2)\hbar/m\mathrm{v}$, the self-consistently determined minimum impact parameter(solid line), obtained by solving

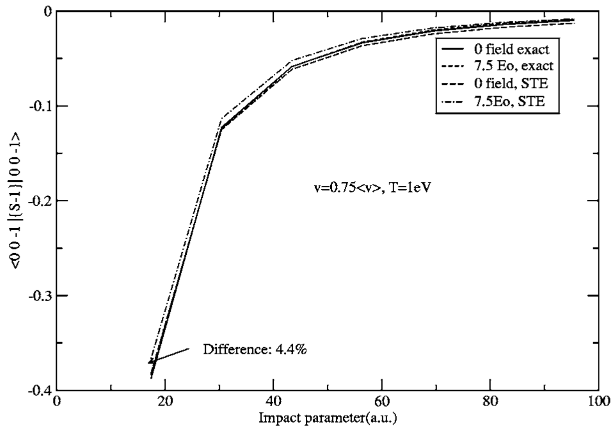

FIGURE 3. Exact vs. STE calculation

$\{I - S_u(\rho,\mathrm{v}\ )S_l^\dagger(\rho,\mathrm{v})\} = I$ (to second order in perturbation theory for the S-matrix product) and the standard(dashed) $(n_u^2 - n_l^2)a_0$ and safe(based on Fig.1)(dot-dashed) WE minimum impact parameter cutoffs, where the subscripts u and l stand for "upper" and "lower" levels respectively. The regime above the solid line is the region of applicability of the dipole perturbative ST. It is clear that the WE cutoff significantly affects the minimum impact parameter determination and furthermore the perturbation is quite strong(since we are significantly below the unitarity-based $\rho_{min}(\mathrm{v})$) at the onset of penetration. Thus for $\mathrm{v} < 1.5\langle\ \mathrm{v}\rangle$ perturbation theory is still not valid at impact parameters $\leq 20a_0$, while for higher velocities the weak collision contribution is overestimated. This measurement will be discussed further along with other experimental data later on.

CALCULATIONS

In the calculations shown in Figs. 3-5 the SE has been solved numerically to obtain the S-matrices required for the exact collision operator. This way all nonperturbative aspects are treated exactly and penetration is included, if the relevant code option is enabled. All calculations reported here are dipole only in order to be able to compare with simplified calculations.

It should be mentioned that such calculations are not new[8]. The main difference is that in the present calculations we also allow for a quasistatic ion field in the calculation of impact broadening. Such a coupling was considered in recent works(e.g. "AGT" mentioned above). Common wisdom[9] holds that it is usually permissible to ignore the electric field in the computation of electron broadening, though not in the final profile. However, as the density increases a progressively larger part of electrons becomes nonimpact and is thus correlated with the static ion field[10]. The issue of a possible electron-ion coupling at high densities is an important one for hydrogen, because if this coupling is unimportant the S-matrix may be solved analytically[11, 12, 13] under

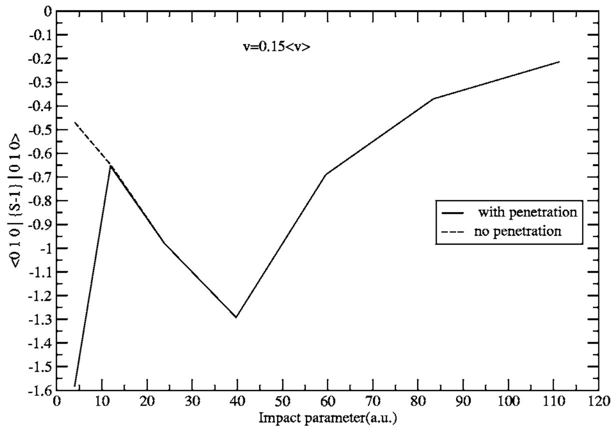

FIGURE 4. Effects of penetration

the assumptions of a pure dipole interaction with no penetration, no-quenching, no fine structure etc. In practice, however, this solution is almost never used in the literature.

All static effects may be treated exactly by using the parabolic basis, i.e. eigenstates of the instantaneous ion field. All calculations refer to $Ly-\alpha$ at 1eV temperature(for which line penetration and nonperturbative effects are the weakest). Within standard assumptions(e.g. electron motion is not affected by the ion field) the only effect of a static field on the broadening operator is to turn hydrogen into a nonhydrogenic species. The corresponding calculation takes into account the finite energy spacings between the states of the upper/lower level and is referred to as STE. The difference from the ST is the imaginary exponentials. Most of the differences between the ST and STE are due to cutoffs. Including these differences, at 1 eV, $10^{18}e/cc$, and even at 5eV and $10^{19}e/cc$, the L_α widths differ by less than 10%. As long as perturbation theory is ok, this is *all* the coupling between electrons and ions. Fig.3 checks STE against exact semiclassical calculations, i.e. the SE is solved and the S-matrix is computed in the parabolic basic for the ion field in the z-direction and the result is integrated over angles describing the orientation of the collision axes with respect to the coordinate frame. The results show that as long as perturbation theory holds, STE is fully satisfactory.

PROBLEMS WITH THE STANDARD THEORY

Problems with the ST arise on two fronts: Neglect of penetration and nonperturbative effects. With respect to the former, the point is that *when* penetration is important, it makes a substantial difference, as Fig.4 illustrates. Fig.5 compares the difference penetration makes against the difference made by the electron-ion coupling.

Even though for L_α penetration is not important, it is still more important than electron-ion coupling. With regard to the later, the best thing ST could do is to compute unitarity-based error bound for the contribution of that part of the phase space(i.e.

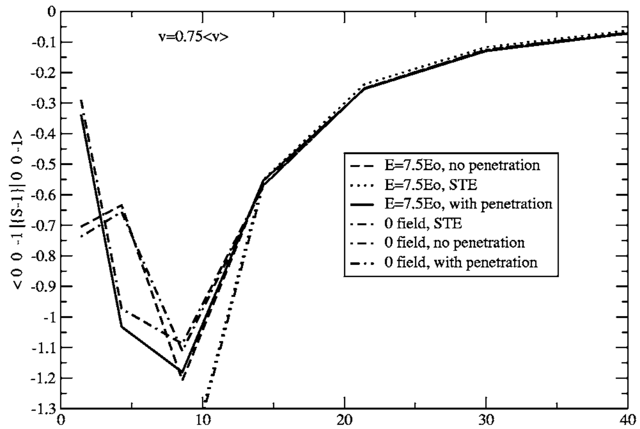

FIGURE 5. Relative importance of electron-ion coupling and penetration

velocities and impact parameters) which could not be computed nonperturbatively. For high densities and low temperatures this part can be comparable to what can be computed presumably reliably by perturbation theory. What is worse, the standard demarcation line between these two regimes was drawn by neglecting the matrix nature and estimating the part involving the dipole matrix elements

$$\langle\alpha|\mathbf{r}|\alpha''\rangle\cdot\langle\alpha''|\mathbf{r}|\alpha'\rangle+\langle\beta'|\mathbf{r}|\beta''\rangle\cdot\langle\beta''|\mathbf{r}|\beta'\rangle-2\langle\alpha|\mathbf{r}|\alpha'\rangle\cdot\langle\beta'|\mathbf{r}|\beta\rangle=(n_u^2-n_l^2)^2a_0^2 \quad (3)$$

In detail this means writing down the expression for the (matrix elements of the) S-matrix as a function of ρ and v (estimating the dipole matrix elements) and equating this to unity. This gives a ρ(v) curve, which is the most sophisticated form of the Weisskopf Radius WR(often the WR is taken to refer to this value at the average velocity). The WR is a fairly naive concept that is often misused in the literature. For instance WR arguments are often used to argue for ions being quasitatic, but the WR is simply a very simple way to check *unitarity*, *not* time scales. What the WR really tells us is that *if* we were to compute the relevant S-matrix elements by perturbation theory, then this would break down at the WR(which is a function of velocity v, though often the average velocity is used) This in turn means that *if* the WR were correct, then collisions with shorter impact parameters would be strong, and the rest weak. This is used in quasistatic approach justifications, essentially saying that strong collisions are static and hence estimating the degree to which a species(usually ions) is static. As pointed out 22 years ago[14], static collisions are strong, but the converse is not true. Consequently using the WR to argue for time scales is wrong and furthermore leads to the absurd result for lines with a central component that at high enough densities electrons too are quasistatic, in which case there is no mechanism left(except Doppler and natural broadening) to broaden the central component, i.e. if we could turn off Doppler and natural broadening, we'd have δ-functions for the central components at very high densities.

In addition to this erroneous, yet still invoked[15] use of the WR, the very calculation of this cutoff is in principle flawed for two reasons: First, one solves a matrix equation as if it were a scalar as in Eq.(3). Second, one computes the second-order S-matrix expansion and believes that unitarity is satisfied if this is small. However, this will be true if the U-matrix has remained small for *all* times(which is true in the case of hydrogen with no static field), which is not always the case for nonhydrogenic species, in which case this perturbative check is fooled[16]. Safe perturbative formulas have since been given[17], that will correctly account for this effect.

OTHER EXOTIC EFFECTS AND COMPARISON WITH EXPERIMENTS

Electron-ion coupling is not the only exotic effect proposed to explain the high density experiments. Other effects include[18] a narrowing effect allegedly caused by the motion of the perturbing electron in the field of the nearest-neighbor ion(the fact that the plasma is electrically neutral and that a nearest-neighbor electron also exists is ignored) as well as a residual ion impact width(among other issues, the fact that at high densities for every ionic configuration that is "impact" there are many more which are "ion dynamic" -and of course even more that are "quasistatic"- is also ignored. Among other things, by assuming that what is not quasistatic is impact, based on the known to be erroneous Weisskopf radius criterion, the dynamic ionic contribution is grossly overestimated, since at a given set of plasma conditions the impact broadening is an upper bound for the possible contribution). By combining these "effects", in the so-called UAGT("Upgraded" AGT) impressive agreement with most high density experimental data is obtained, for instance Fig.1 in Ref.[18]. It is interesting however, to compare this agreement with ref.[5] which is essentially an *exact* dipole calculation, that essentially confirms the results of the ST(roughly correctly reproduced in that figure, except that error bounds are not given). As proposed in ref.[5] and confirmed by the experimental group[18] the original lowest density points were affected by self-absorption. The exact dipole calculation in [5] *agrees* with the new measurement around $2.4\text{x}10^{18}$e/cc (the only point common in the original and new measurements) *without* invoking any of these additional effects, as well as with the highest density measurements within the theoretical error bars, which arise from the need to preserve the dipole and semiclassical approximations(not unitarity, which is satisfied automatically in the exact dipole calculations). With regard to the dipole approximation the $n_u^2 a_0$ cutoff was used in [5]. The results in [18] have much lower error bars because they simply *ignore* these considerations and pretend one can integrate down to 0 impact parameters.

We should stress that these additional "effects" are quite important for achieving "agreement" with experiment. For instance in [4] let's look at the point at T=20500K and density $2.02\text{x}10^{18}$e/cc. Roughly one fourth of the UAGT width is a residual ion impact width. The ST is in agreement with the FFM and MMM results. A self-consistent(i.e. with self-consistent unitarity-preserving $\rho_{min}(\text{v})$ determination) gives an STE width almost twice as large as the ST. However, these values are computed with a minimum impact parameter also larger than the de Broglie wavelength $\hbar/m\text{v}$ and WE $(n_u^2 - n_l^2)a_0$, and a strong collision contribution based on the assumption of a constant $\{I - S_u(\rho,\text{v})S_l^\dagger(\rho,\text{v})\} = 1$ for strong collisions. When we replace this WE cutoff by $18a_0$(as obtained from Fig.1), the weak collision width is 49Å in the ST and 32Å in the STE case, i.e. the region for which penetration is important(and which may not be accurately computed by either the ST without penetration or the more exotic approaches which again ignore these issues) is responsible for a large part of the width. In short, all these approaches are insufficient for comparing with experiments in that regime.

CONCLUSIONS

Recent high density experiments with hydrogen lines can be important for checking the contributions of close collisions(in particular for testing the adequacy of a semiclassical treatment of close collisions down to the de Broglie wavelength, which is a theoretically important issue[1]. To do so, however, requires a theoretical analysis that takes into account important physics such as penetration and nonperturbative aspects. Thus far there is *no* evidence against the adequacy of the standard electron impact broadening picture, provided close collisions are treated accurately. Exotic effects proposed to explain the apparent discrepancies between a dipole nonperturbative approach and experiment are clearly incorrect, both in theory and in terms of their results.

REFERENCES

1. S.Alexiou, and R.W.Lee, "Electron Line Broadening in Plasmas: Resolution of the Quantum vs. Semiclassical Calculation Puzzle," in *Spectral Line Shapes*, edited by J.Seidel, AIP Conference Proceedings 559, American Institute of Physics, New York, 2001, pp. 135–143.
2. St.Böddecker, S.Günter, A.Könies, L.Hitzschke, and H.-J.Kunze, *Phys.Rev.***E**, **47**, 2785–2791 (1993).
3. A.Escarguel, B.Ferhat, A.Lesage, and J.Richou, *JQSRT*, **64**, 353–361 (2000).
4. S.Flih, and Vitel, Y., "Experimental Profiles of Hydrogen Balmer α Line Emitted in weakly Non-ideal Plasma," in *Spectral Line Shapes*, edited by J.Seidel, AIP Conference Proceedings 559, American Institute of Physics, New York, 2001, pp. 30–32.
5. S.Alexiou, and E.Leboucher-Dalimier, *Phys.Rev.E*, **60**, 3436–3438 (1999).
6. Touma, J., Oks, E., Alexiou, S., and A.Derevianko, *JQSRT*, **65**, 543–571 (2000).
7. G.Hegerfeld, and V.Kesting, *Phys.Rev. A*, **37**, 1488–1496 (1988).
8. K.Bacon, K.Y.Shen, and J.Cooper, *Phys.Rev.*, **188**, 50–56 (1969).
9. Baranger, M., *Atomic and Molecular Processes*, Academic, New York, 1962, p. 523.
10. Alexiou, S., *Phys.Rev.Lett.*, **76**, 1836–1839 (1996).
11. H.Pfennig, *JQSRT*, **12**, 821–837 (1972).
12. H.Pfennig, *Z.Naturforsch.a*, **26**, 1071– (1971).
13. V.S.Lisitsa, and G.V.Sholin, *Sov.Phys.JETP*, **34**, 484–489 (1972).
14. J.Seidel, "Theory of Hydrogen Stark Broadening," in *Spectral Line Shapes*, edited by B.Wende, W. de Gruyter, Berlin, 1981, pp. 3–40.
15. E.Oks, "Dramatic Effect of the accelaration of electrons by the ion field on spectral line shifts in plasmas: advanced theory," in *these proceedings*, 2003.
16. S.Alexiou, *Phys.Rev.Lett.*, **75**, 3406–3409 (1995).
17. A.Poquérusse, S.Alexiou, and E.Leboucher-Dalimier, *JQSRT*, **56**, 797 (1997).
18. E.Oks, "Comparison of the Latest Experimental H-alpha Results at the Bochum's Gas-Liner Pinch with the Upgraded Advanced Generalized Theory," in *Spectral Line Shapes*, edited by J.Seidel, AIP Conference Proceedings 559, American Institute of Physics, New York, 2001, pp. 54–57.